Evolving but not solving: A Genetic Algorithm's struggle with LLM reasoning about an introductory physics problem

CAL POLY

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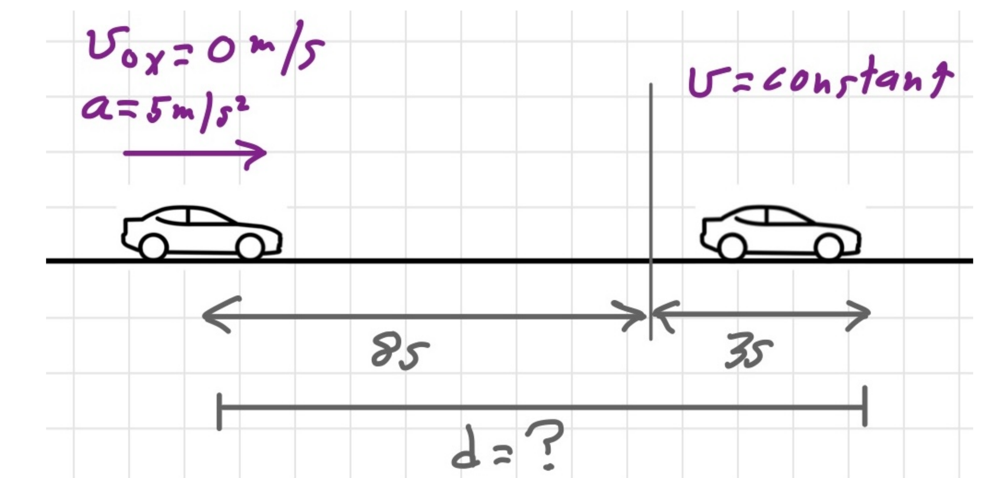
Summary

Can a Large Language Model (LLM) play the role of a student in a Socratic (Q&A-style) tutorial session with an expert, to solve an introductory-level physics problem?

The problem



A car at rest accelerates from a traffic light at $5\ m/s^2$ for $8\ s$, then coasts for 3 more seconds. How far will it travel?

Needed logic

- Two acceleration zones to be identified
- Zones must be ordered sequentially
- Kinematics from earlier zone must feed later zone
- Proper use of physics equations

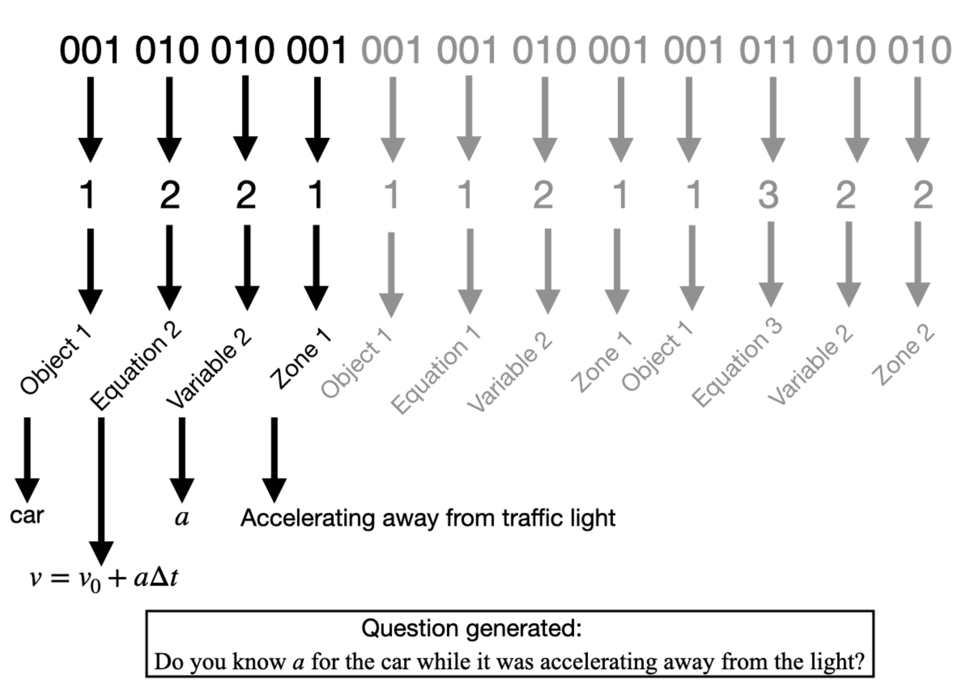
The expert: an optimization algorithm

Any 1D kinematics problem can be solved with 3 equations. An algorithm generates questions based on them.

Equation Number	Equation	Unknowns
1	$\overset{①}{x} = \overset{②}{x_0} + \overset{③}{v_{0x}} \overset{④}{\Delta t} + \frac{1}{2} \overset{⑤}{a_x} \Delta t^2$	5
2	$\overset{①}{v_x} = \overset{②}{v_{0x}} + \overset{③}{a_x} \overset{④}{\Delta t}$	4
3	$\overset{①}{\Delta t} = \overset{②}{t} - \overset{③}{t_0}$	3

A Genetic Algorithm (GA): The automated expert

As described[1], a GA forms tuples binary strings in the form of *(object,equation,variable,acceleration zone)*, and compiles a question from the tuple.



The GA seeks an optimal path to a solution based on answers to the questions that dictate discovered knowledge about the problem (x, x_0, v, v_0, a and Δt).

SAMPLE DIALOG

```
Do you see any objects in the problem? y
What do you see? a car
Do you see any more objects in the problem? n
Do you know v0x for the a car? y

O.K., you are claiming to  know v0x for the a car
Ok, what is v0x then? 0 m/s
What the a car was doing when it had the speed of the a car before it started (which you
said was 0 m/s)? starting at the traffic light
Do you know ti for the a car? n
Do you know ax for the a car? y

You are claiming to  know ax for the a car
But be sure that:
ax is the acceleration a car had after starting with speed v0x, which you said was 0 m/s
(the speed of the a car before it started starting at the traffic light).  Is this the
right scenario for the ax you now claim to know? [y/n]: y
Ok, what is ax then? 5 m/s^2
Do you know dt for the a car? y

You are claiming to  know dt for the a car
But be sure that:
dt is the time interval that started when a car had a speed of v0x, which you said was 0
m/s (the speed of the a car before it started starting at the traffic light)
dt is the time interval during which a car had an acceleration of ax, which you said was 5
m/s^2 (the acceleration of the a car starting at the traffic light). Is this the right
scenario for the dt you now claim to know? [y/n]: y
Ok, what is dt then? 8 s
Do you know x for the a car? n
Do you know any other ax for the a car? y
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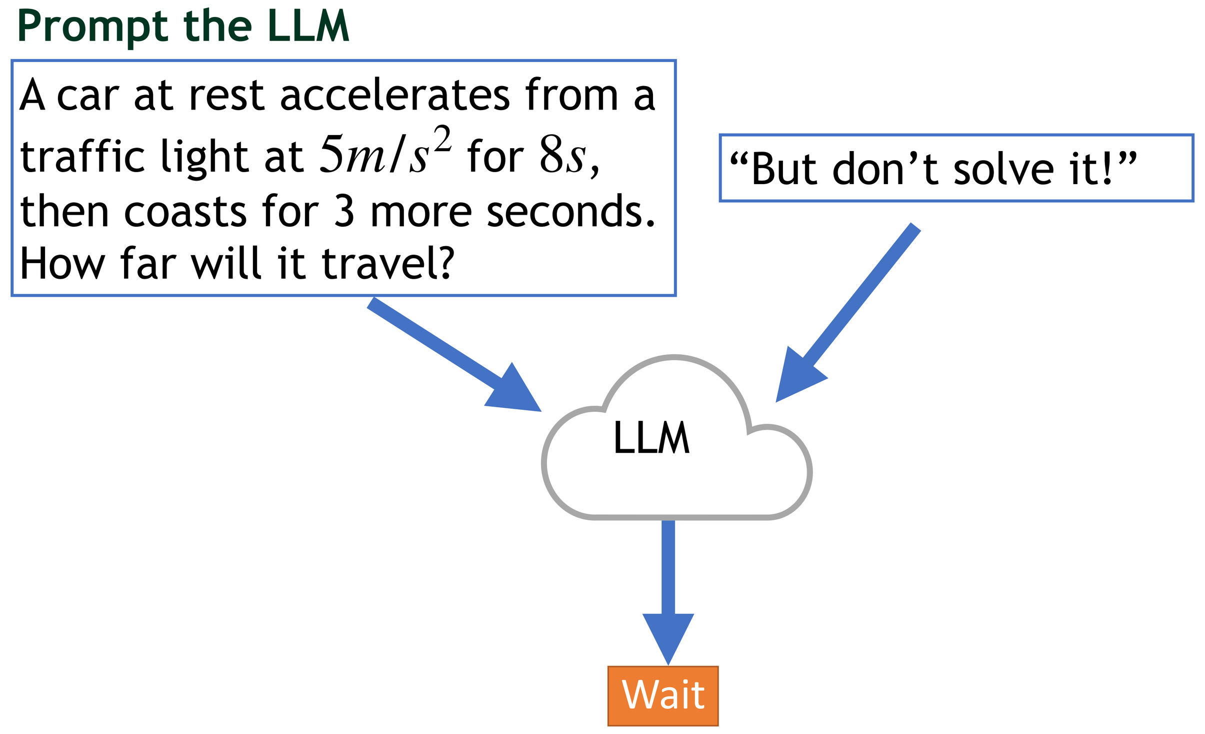
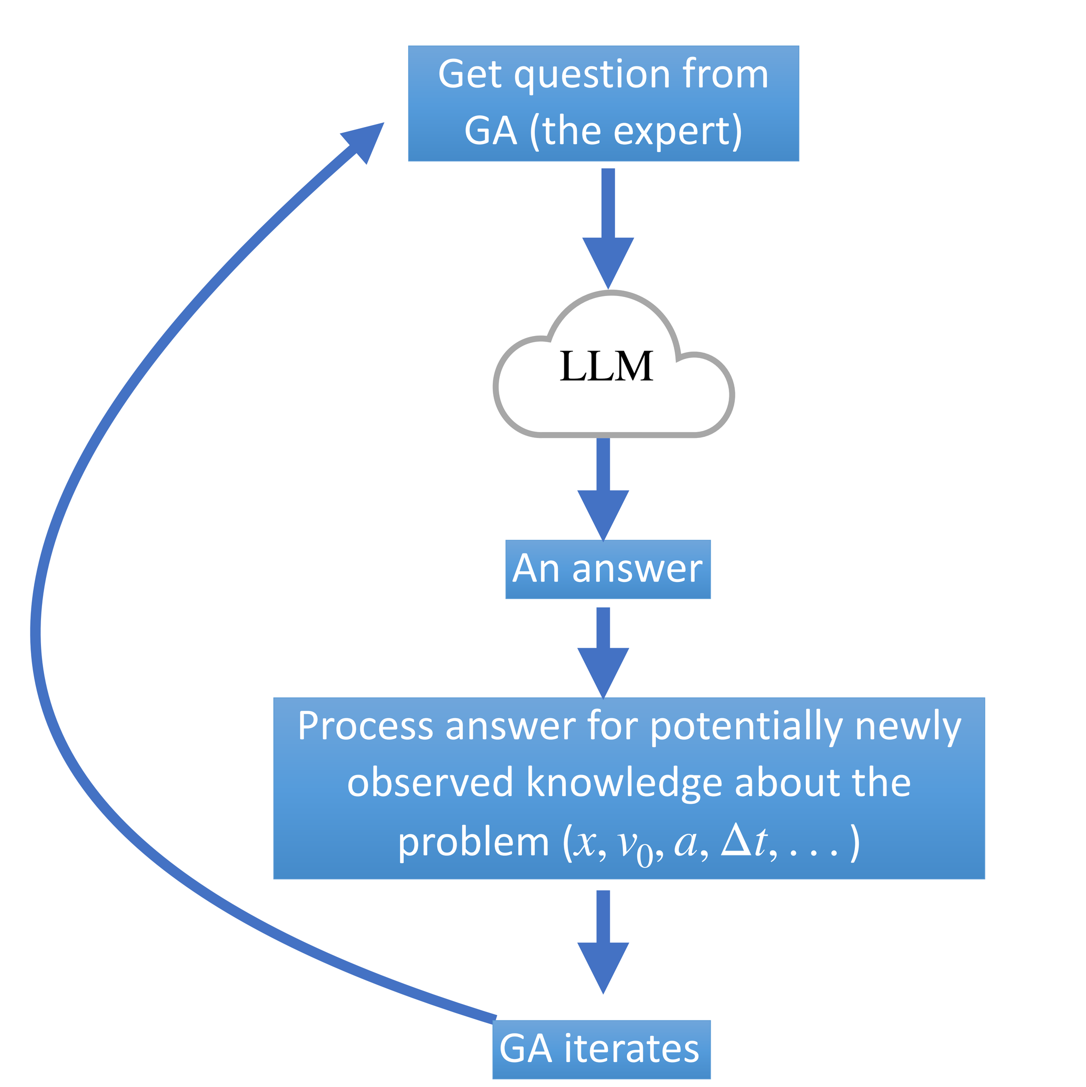
A sample dialog between the GA-generated questions and a human (see Ref. 1).

An LLM as the student

Prompt the LLM





Results

LLM successes

- Recognizes the object (car)
- Recognizes two acceleration zones: "it increases velocity" and "no change in velocity."
- Properly orders zones temporally: 1: "it increases velocity", 2: "no change in velocity."

Typical equation sequence found by the LLM

Answer

x_0 for zone #1?

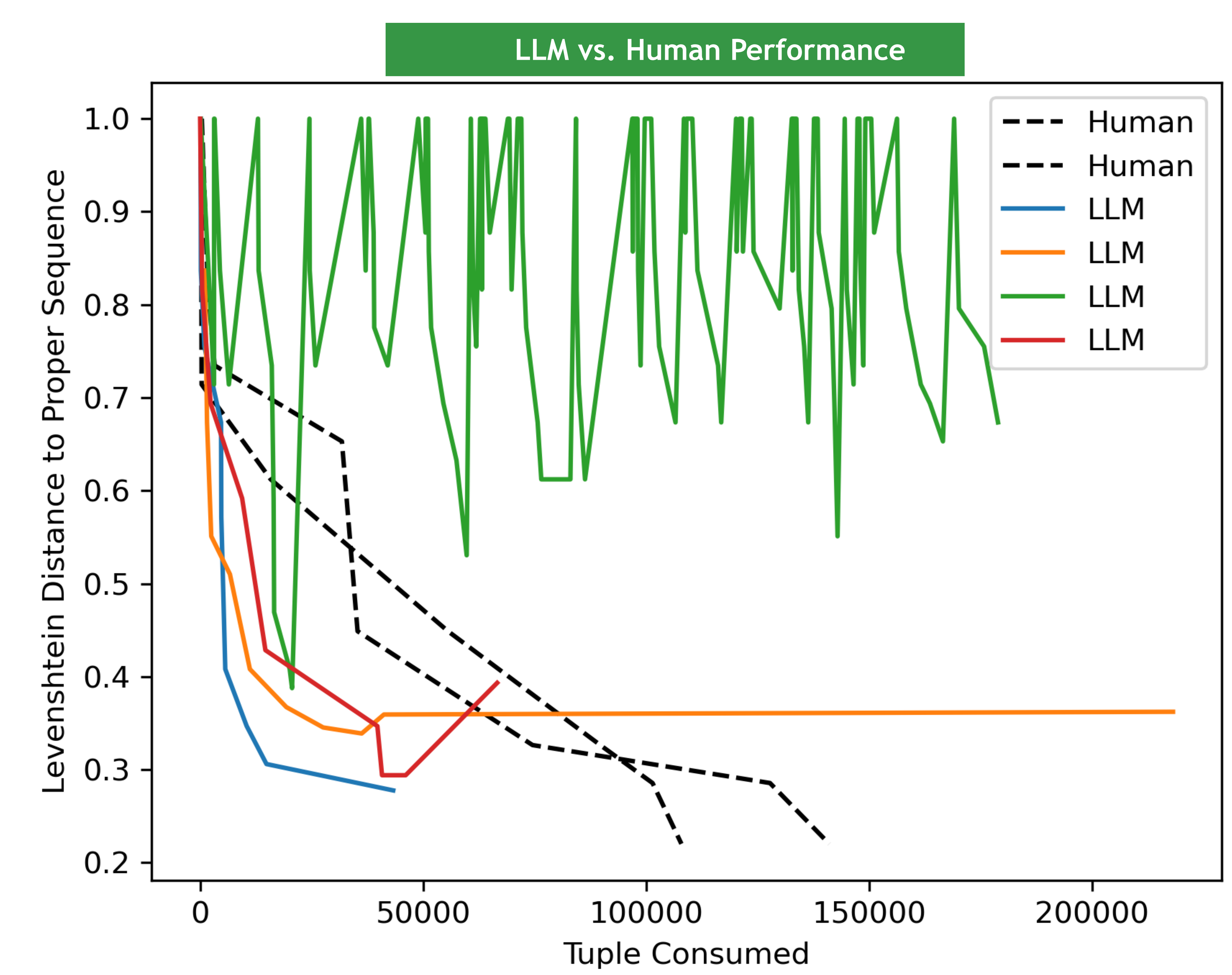
- Sequence found: [{'x': 1}, {'ax': 2}, {'ax': 1}, {'dt': 1}, {'vx': 2}, {'dt': 2}, {'v0x': 2}, {'x': 2}, {'vx': 1}, {'v0x': 1}, {'x0': 2}]

Optimal human generated sequence

- A proper sequence: [{'dt': 1}, {'x0': 1}, {'v0': 1}, {'a': 1}, {'dt': 1}, {'a': 2}, {'x': 2}, {'v': 5}, {'x0': 4}, {'v0': 4}, {'x': 4}]

Overall LLM assessment

- Sloppy work with illogical parameter discovery
- Cannot seem to associate terminal values of zone #1 with initial values of zone #2
- Cannot seem to work with strict meaning of kinematic variables: x_0 vs x and v_0 vs v, and ties between them.
- Likes to skip steps. do $x = \frac{1}{2}(5m/s^2)(8s)^2 = 160$ m and $v = (5m/s^2)(8s) = 40m/s$ then $x = (40m/s)(3s) = 120$ m



Results of multiple LLM and human attempts to reason through the problem. The Levenshtein Distance between the LLM-produced equation sequence versus the optimal (human made) sequence, as a function of tuples needed from the GA. (0=best, 1=worst). The human-led session (dashed) solves the problem with 100% efficient in its answers and use of logic. The LLM often gets stuck (green) or finds an answer sooner by skipping important logical steps (blue/red).

Conclusions

Answer to opening question: "no"

The LLM cannot seem to maintain a logical presence to guide itself to a solution or match the human performance.

The LLM does not seem to like the persistent stream of questions. Common occurrences:

- Recites poetry
  "She walked on trees as if they were roads made of bark and wind, her small feet gliding over roots that twisted like ribbons of green"
- Strange responses
  "**… … ……………..….. …**…"
- Talks to itself
  "But we should not use "over time"? But we can keep it short: "It increases speed." That's okay."
- Gets frustrated
  "The user is continuing a back-and-forth. We need to comply with the instructions."
- Defends itself
  Expert: "You made a mistake. There's only two different areas of acceleration in this problem, not 3. Can you describe why you think there's more than 2 acceleration zones in this problem?

  LLM: "I thought the start of coasting (the instant the acceleration changes from 5 m/s² to 0 m/s²) might count as a separate zone, and then the coast itself as another zone. However, that instant is not a distinct acceleration period, so there are only two zones: acceleration and zero acceleration."

REFERENCES

1 Tom Bensky and Justin Kopcinski, "Use of a genetic algorithm to find solutions to introductory physics problems", arXiv:2508.10920 [cs.NE].

2 C.F. Tsai, et. al, "Can Large Language Models Play Text Games Well? Current State-of-the-Art and Open Questions", arXiv:2304.02868 [cs.CL]